Roll No......

END SEMESTER EXAMINATION DEC-2024

Name of the Program: B. Tech Petroleum Engineering
Name of the Course: Petroleum Drilling Engineering I

Semester: 3rd
Course Code: TPE 304

Time: 3 Hrs

Maximum Marks: 100

Notes:

(i) All questions are compulsory.

(ii) Answer any two sub questions among a,b, & c in each main questions

(iii) Total marks in each main question are twenty.

(iv) Each questions carries 10 marks.

Q1	(20 marks)	
(a)	Describe the component and purope of hosting system of a rotary drilling rig with the help of diagram.	CO1
(b)	What are primary and secondary well control? List and describe the common reasons for loss of primary control	CO1
(c)	Describe the procedure for controlling a kick when drilling and when tripping.	CO1
Q2	(20 marks)	
(a)	What is a Ton-Mile? Discuss the Importance of Ton-Miles in Drilling Operations	CO1
(b)	Describe the functions of drilling mud.	CO3
(c)	Discuss the advantages of using the reciprocating positive displacement pump over centrifugal pump.	CO3
Q3	(20 marks)	
(a)	What is the purpose of casing? List and describe the different types of casing. What is the difference between casing and liner ?	CO4
(b)	What is the purpose of cementing? Explain primary cementiing, Squeeze cementing and plug back cementing.	CO4
(c)	Explain cement bond logging.	CO4
Q4	(20 marks)	
(a)	Define the terms: Pressure Gradient; hydrostatic pressure; "Normal" Pressure; "Abnormal" Pressure; Overburden(geostatic) Pressure; Fracture pressure.	CO5
(b)	Describe in general terms the origins and mechanisms which generate Overpressured and Underpressured reservoirs	CO5
(c)	Describe the impact of abnormally pressured formations on drilling operations	CO5
Q5	(20 marks)	
(a)	Describe the main features of roller cone bit. Why the roller cone bit is prefered over the drag bits.	CO6

(b)	List and describe the factors affceting penetration rate of bit.	CO6
(c)	Rock sample undera 2000-psi confining pressure fails when subjected to compressional loading of 10000 psi along plane which makes an angle of 27 with the direction of the compressional load. Using the Mohr failure criterion determine the angle of internal friction the shear strength and the cohesive resistance of the material.	CO2